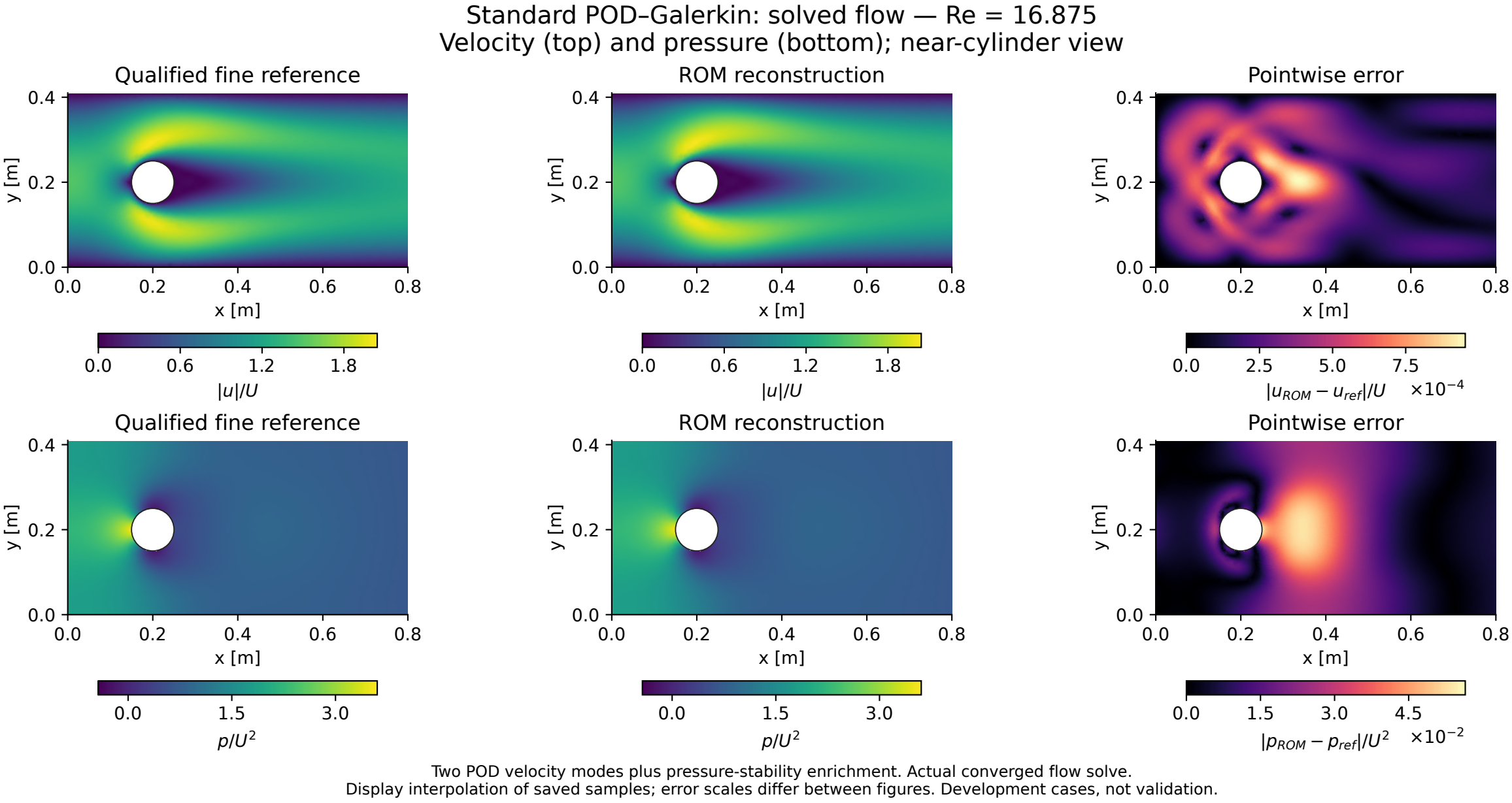

Standard POD–Galerkin: solved flow — Re = 16.875
Velocity (top) and pressure (bottom); near-cylinder view
Qualified fine reference
ROM reconstruction
Pointwise error
y [m]
x [m]
0.0
0.2
0.4
0.6
0.8
0.0
0.6
1.2
1.8
|u|/U
0.0
2.5
5.0
7.5
|u_ROM − u_ref|/U
×10^−4
Qualified fine reference
ROM reconstruction
Pointwise error
0.0
1.5
3.0
p/U^2
0.0
1.5
3.0
4.5
|p_ROM − p_ref|/U^2
×10^−2
Two POD velocity modes plus pressure-stability enrichment. Actual converged flow solve.
Display interpolation of saved samples; error scales differ between figures. Development cases, not validation.